

Chapter 5: Differentiation

Section: Derivatives of Higher Order

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Definition

When Does the n th Derivative Exist?

Summary

Motivation

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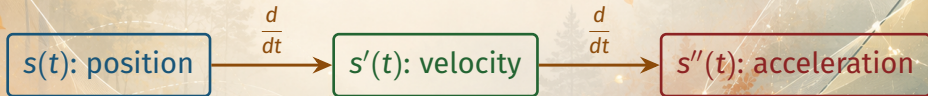
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- Iterating gives $f'', f^{(3)}, \dots, f^{(n)}$ — the raw material of **Taylor's theorem** (next episode)

A Familiar Instance: Motion

Position, velocity, acceleration

If $s(t)$ is the position of a particle at time t , then s' is its **velocity** and s'' is its **acceleration**.





The Definition

Definition 5.14: Higher Order Derivatives

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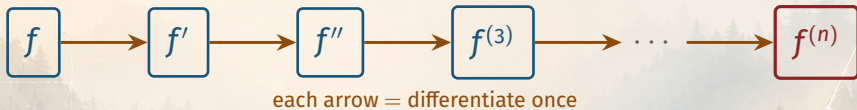
Definition 5.14 (continued)

Continuing in this manner, we obtain functions

$$f, f', f'', f^{(3)}, \dots, f^{(n)},$$

each of which is the derivative of the preceding one. $f^{(n)}$ is called the **n th derivative**, or the **derivative of order n** , of f .

The Tower of Derivatives



- Each stage must exist before the next one
- The chain may continue forever — or **stop**

Example: A Polynomial Never Stops

Example: $f(x) = x^3$

$$f'(x) = 3x^2, \quad f''(x) = 6x, \quad f^{(3)}(x) = 6, \quad f^{(4)}(x) = 0, \quad f^{(n)}(x) = 0 \quad (n \geq 4)$$

Derivatives of **every** order exist, at every point.

More generally

Every polynomial of degree d has derivatives of all orders;

$$f^{(n)} \equiv 0 \text{ for all } n > d.$$

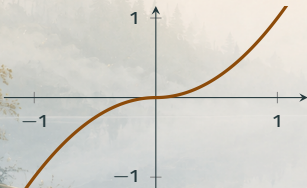
Example: The Tower Can Stop

Example: $f(x) = x|x|$

$f'(x) = 2|x|$ exists for **every** x , but f' has a **corner** at 0.

So $f''(0)$ does **not** exist.

$$f(x) = x|x|$$



$$f'(x) = 2|x| \quad (\text{corner at } 0)$$





When Does the n th Derivative Exist?

The Fine Print of Definition 5.14

For $f^{(n)}(x)$ to exist at a point x ...

- $f^{(n-1)}(t)$ must exist for all t in a **neighborhood** of x (one-sided if x is an endpoint of the interval)

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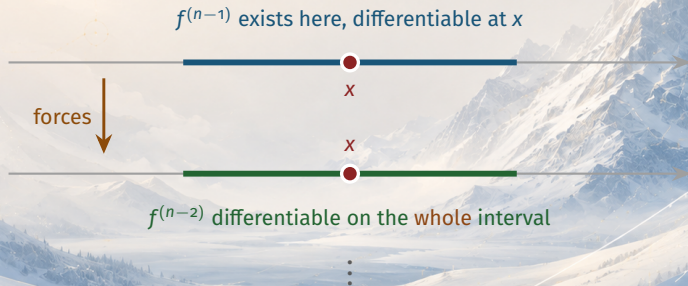
- $f^{(n-1)}(t)$ must exist for all t in a **neighborhood** of x (one-sided if x is an endpoint of the interval)
- $f^{(n-1)}$ must be **differentiable at x** :

$$f^{(n)}(x) = \lim_{t \rightarrow x} \frac{f^{(n-1)}(t) - f^{(n-1)}(x)}{t - x}$$

The Requirement Cascades Down the Tower

Cascade

Since $f^{(n-1)}$ must exist in a neighborhood of x , $f^{(n-2)}$ must be **differentiable in that whole neighborhood** — and so on, down the tower.



The background is a digital illustration of a vast, snowy mountain range under a cloudy sky. Overlaid on this scene are intricate golden geometric patterns, including circles, arcs, and lines connecting points, resembling a celestial or mathematical diagram. The word "Summary" is centered in the upper half of the image.

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Key takeaways from this section:

- **Definition 5.14:** differentiate repeatedly to get the tower $f, f', f'', f^{(3)}, \dots, f^{(n)}$; each function is the derivative of the preceding one

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- **Existence is local and cascading:** $f^{(n)}(x)$ requires $f^{(n-1)}$ on a **neighborhood** of x , hence $f^{(n-2)}$ differentiable there, and so on

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- **Definition 5.14:** differentiate repeatedly to get the tower $f, f', f'', f^{(3)}, \dots, f^{(n)}$; each function is the derivative of the preceding one
- **Existence is local and cascading:** $f^{(n)}(x)$ requires $f^{(n-1)}$ on a neighborhood of x , hence $f^{(n-2)}$ differentiable there, and so on
- **The tower may stop:** $x|x|$ has f' everywhere but no $f''(0)$; polynomials have derivatives of all orders

Coming up next: Taylor's Theorem (5.15) — approximating f by a polynomial built from $f(x), f'(x), \dots, f^{(n-1)}(x)$.